

Hand-in assignment 4 solutions

Please note that these solutions are only meant to serve as a guide, the questions may have a number of alternative acceptable solutions. The advise is: **If you do not get full mark for a question, try to do it again independently after going through this document.**

Solutions

In this document, d_E and d_D denotes the usual metric and discrete metric, respectively on the sets they are considered.

1. (a) False. Consider metric space $(\mathbb{R}, d_E)$ and a family of compact subsets $(A_n)_{n \in \mathbb{N}}$ given by $A_n = [0, 1 - \frac{1}{n}]$, their union $\bigcup_{n \in \mathbb{N}} A_n = [0, 1)$ is not compact.
 - (b) False. Consider metric space $(\mathbb{R}, d_E)$ and the empty family of compact subsets, then $\bigcap \emptyset = \mathbb{R}$ is not compact.
 - (c) False. Consider metric space $(\mathbb{R}, d_E)$ and a function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 0$ for each $x \in \mathbb{R}$, f is continuous and $\{0\}$ is compact, however $f^{-1}(\{0\}) = \mathbb{R}$ is not compact.
 - (d) False. Consider metric space $(\{0, 1\}, d_D)$, subsets $\{0\}$ and $\{1\}$ are connected, however, $\{0\} \cup \{1\} = \{0, 1\}$ is not connected.
 - (e) False. Consider metric space $(\mathbb{R}^2, d_E)$, subsets $A_1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ and $A_2 = \{(x, y) \in \mathbb{R}^2 \mid (x - 1)^2 + y^2 = 1\}$ are connected, however, their intersection $A_1 \cap A_2 = \{(\frac{1}{2}, \frac{\sqrt{3}}{2}), (\frac{1}{2}, -\frac{\sqrt{3}}{2})\}$ is not connected.
2. ($\implies$) Suppose (A_n) is a decreasing sequence of non-empty subsets of X .
 For each $n \in \mathbb{N}$, take $a_n \in A_n$. (This is possible because $A_n \neq \emptyset$.)
 By assumption, there exists a subsequence (a_{n_k}) of (a_n) such that $a_{n_k} \rightarrow x$ for some $x \in X$.
 We show that $x \in \bigcap_{n \in \mathbb{N}} \overline{A_n}$.
 Suppose U is an open set containing x .
 Take $N \in \mathbb{N}$ such that $k \geq N$ implies that $a_{n_k} \in U$.
 Let $m \in \mathbb{N}$.
 Take $k \in \mathbb{N}$ such that $k \geq N$ and $n_k > m$.
 Then $A_m \supseteq A_{n_k}$, so $U \cap A_m \neq \emptyset$.

($\impliedby$) Suppose (x_n) is a sequence in (X, d) .

For $N \in \mathbb{N}$, define $T_N = \{x_m : m \geq N\}$.

Then (T_n) is a decreasing sequence of non-empty subsets of X , so by assumption, $\bigcap_{n \in \mathbb{N}} \overline{T_n} \neq \emptyset$.

Take $x \in \bigcap_{n \in \mathbb{N}} \overline{T_n}$.

We find a subsequence of (x_n) that converges to x .

Let $x_{n_1} \in B(x; 1) \cap T_1$. (This is possible because $x \in \overline{T_1}$.)

Let $x_{n_2} \in B(x; \frac{1}{2}) \cap T_{n_1+1}$.

$\vdots$

Let $x_{n_k} \in B(x; \frac{1}{k}) \cap T_{n_{k-1}+1}$.

$\vdots$

Then (x_{n_k}) is a subsequence of (x_n) that converges to x as $k \rightarrow \infty$.